Hareshkumar Jadav

hareshjadavcse@gmail.com Website linkedin Google Scholar

Education

PhD **Indian Institute of Technology Indore**, Computer Science — Jul 2023 – Present
- CGPA: 9.24/10
- Supervisor: Dr. Ranveer Singh, IIT Indore, India
- Co-Supervisor: Dr. Sudipta Mallik, Marshall University, USA

M.Sc **The Maharaja Sayajirao University of Baroda**, Mathematics — Aug 2020 – May 2022
- CGPA: 8.1/10

B.Sc **Dr. Subhash Science College Junagadh** , Mathematics — Jul 2017 – Apr 2020
- CGPA: 7.9/10

Research Interest

Combinatorial optimization, Algorithm, Algebraic Graph Theory

Experience

Visiting Scholar — Jul 2025 - Oct 2025
- School of Industrial Engineering, Purdue University, USA
  Project Supervisor: Prof. Vaneet Aggarwal
  Project title: Stronger Approximation Guarantees for Non-Monotone γ-Weakly DR-Submodular Maximization

Junior Research Fellow — Nov 2022 - June 2023
- Computer Science and Engineering Department, IIT Indore
  Project Supervisor: Dr. Ranveer Singh
  Project title: The Construction of Expander and Ramanujan Graphs and Their Application in Robust Network topologies

Publications

Theoretical Machine learning

Upper-Linearizability of Online Non-Monotone DR-Submodular Maximization over Down-Closed Convex Sets — Feb 2026
Yiyang Lu, *Hareshkumar Jadav*, Mohammad Pedramfar, Ranveer Singh, Vaneet Aggarwal
ICML 2026, link

Stronger Approximation Guarantees for Non-Monotone γ-Weakly DR-Submodular Maximization — May 2026
Hareshkumar Jadav, Ranveer Singh, Vaneet Aggarwal
AAMAS 2026, link, TMLR link

A Unified Reduction for Delayed Online Maximization of Structured Non-Convex Functions — Under review AAAI 2027
Hareshkumar Jadav, Yiyang Lu, Mohammad Pedramfar, Ranveer Singh, Vaneet Aggarwal

Online Convex Optimization with Dueling Feedback — Under review ICLR 2027
Yiyang Lu, *Hareshkumar Jadav*, Mohammad Pedramfar, Ranveer Singh, Vaneet Aggarwal

A Unified Framework for Analyzing Meta-Algorithms in Riemannian Online Convex Optimization Under review AISTATS 2027

Rudra Vavadiya, *Hareshkumar Jadav*, Ranveer Singh, Vaneet Aggarwal

Graph Theory

A construction of random bigraphs and their application to error correction codes Jan 2025

Hareshkumar Jadav, Ashita Gupta, Ranveer Singh

Journal of Statistical Mechanics: Theory and Experiment, link

Strengthening Wilf's inequality on clique number Jul 2025

Hareshkumar Jadav, Sreekara Madyastha, Rahul Raut, Ranveer Singh

ISSAC 2025, link

Cartesian Prime Graphs and Cospectral Families Jul 2025

Abhinav Bitragunta,*Hareshkumar Jadav*, Ranveer Singh

ISSAC 2025, link

Generating Constrained Lattice Paths in a Grid related to counting cycles Sept 2025

Hareshkumar Jadav,Mihir Patel, Samip Shah, Ranveer Singh, Harsh Talati

ICTCS 2025, link

Moore–Penrose Inverse of Incidence Matrices of Unicyclic Graphs Under review

Hareshkumar Jadav, Priyanshu Pant, Ranveer Singh, Virendra Singh

Moore–Penrose Inverses of Laplacians of Unicyclic Graphs Under review

Hareshkumar Jadav, Priyanshu Pant, Ranveer Singh, Virendra Singh

Stronger Spectral Bound for Clique Number Under review

Hareshkumar Jadav, Sudipta Mallik, Akhilesh Panchangam, Ranveer Singh

Achievements & Awards

- Awarded support of **USD 5000** to attend the **October 2026 Iliad Intensive** at San Francisco, USA, 2026.
- **Best Teaching Assistant Award** for the course **Algebraic Graph Theory, Fall 2026**, presented by the Department of Computer Science and Engineering, IIT Indore. Selected from among 120 teaching assistants in the department.
- **Second prize in poster presentation** on National Mathematics Day at IIT Indore, 2026, for **presenting research on Ramanujan Graphs**.
- **Selected for short-term collaborative research fellowship** 2024-25 to visit Prof. Vaneet Aggarwal's CLAN Labs for three months at **Purdue University, USA**, Funded by IIT Indore.
- Received **Travel Support** from IIT Indore to attend the 26th Conference of the International Linear Algebra Society, held from June 23–27, 2025, in Kaohsiung, Taiwan.
- Qualified GATE (Computer Science) in February 2023.
- Qualified CSIR-NET (Mathematical Science) in September 2022 with All India Rank-61.
- Qualified CSIR-NET (Mathematical Science) in February 2022 with All India Rank-47.
- Qualified Gujarat State Eligibility Test (GSET) in Mathematical Science (2021)

Academic Responsibilities

- Member, Events & Outreach Team, Department of Computer Science and Engineering, IIT Indore, 2024.
- Member, Organizing Team, Workshop on *Ramanujan Graphs*, held at IIT Indore, October 16, 2024.
- Member, Organizing Team, High-End Workshop on *Mathematical Foundation of Machine Learning and Its Application to Emerging Areas of AI*, held at IIT Indore, May 27–June 1, 2024.
- Member, Organizing Team, Advance Instructional School on Algorithmic Graph Theory, held at IIT Indore, June 26–July 15, 2023.

Workshop & Conference

- Presented a poster at the IEEE Information Theory Society (IEEE ITSoc) Summer School 2026, held at IIT Delhi, June 15–19, 2026.
- Presented a poster at the 40th Annual Conference of the Ramanujan Mathematical Society, held at IIIT Delhi, December 18–20, 2025.
- Attended the 26th Conference of the International Linear Algebra Society, held in Kaohsiung, June 23–27, 2025.
- Attended the Summer School on Approximation Algorithms 2025, held at IISc Bangalore, May 21–25, 2025.
- Attended ACM India Summer School on Algorithmic Approaches to NP-Completeness, held at PSG College of Technology, Coimbatore, July 15–26, 2024.
- Attended Mathematical Foundation of Machine Learning and its Application to the Emerging Area of AI, held at IIT Indore, May 27–June 1, 2024.
- Attended High-End Workshop on Mathematical Foundations for AI/ML at IIT Indore May 27 - June 1, 2024.
- Attended Spectral Method in Algorithms, FSTTCS 2023 Workshop, held at IIIT Hyderabad, December 21–22, 2023.
- Attended Advance Instructional School on Algorithmic Graph Theory, held at IIT Indore, June 26–July 15, 2023.
- Attended Advance Instructional School on Combinatorics and Graph Theory, held at the University of Kashmir, May 1–13, 2023.
- Attended CREST School: Applied Combinatorics, Cryptography and Combinatorial Topology, held at TCG CREST, Kolkata, July 11–22, 2022.
- Attended the State Level Workshop on Problem Solving and Enrichment in Mathematics, December 24–27, 2019.

Programming skills:

C/C++, Python, LaTeX, Matlab, SageMath

References

- **Dr. Ranveer Singh**
  Associate Professor, Department of Computer Science, IIT Indore, India
  Email: ranveer@iiti.ac.in
- **Prof. Vaneet Aggarwal**
  Professor, School of Industrial Engineering, Purdue University, USA
  Email: vaneet@purdue.edu
- **Dr. Sudipta Mallik**
  Assistant Professor, Department of Mathematics and Physics, Marshall University, USA
  Email: mallik@marshall.edu